A Consensus-Trained U-Net with a Learned Component Scorer for Solar Filament Instance Segmentation, and What Limits Panoptic Quality on MAGFiLO

Mukharbek Organokov
Independent researcher
mukharbek.organokov@gmail.com

Abstract—We segment solar filaments on GONG H-alpha full-disk images of the MAGFiLO dataset with a single U-Net (ResNet-34 encoder) at 1536 px, trained on the consensus of the annotators and averaged over four flips. Its connected components are candidates; a gradient-boosted component scorer, trained on out-of-fold candidates, predicts each one's expected contribution to Panoptic Quality (PQ) and keeps those that pay. On all 707 training images, scored out of fold with a split by calendar month, the pipeline reaches PQ 0.422 (segmentation quality 0.674, recognition quality 0.626), against 0.409 for the same network at 1024 px with a hand-tuned rule; the public leaderboard gives 0.35. One MAGFiLO annotator scored against another on the same images reaches only PQ 0.340. We report the experiments that did not help (a Mask R-CNN detector, a crop refiner, a larger encoder, limb-darkening correction, a Lovász loss, ensembling, 2048 px input and seven post-processing rules) and show that most of the remaining loss is annotator disagreement rather than model error.

I. Introduction

Filaments are dense, cool plasma suspended above polarity inversion lines; they appear as dark, elongated structures in H-alpha and are at the origin of many eruptions. MAGFiLO [2] provides manual segmentations of filaments on GONG H-alpha images, with up to three independent annotators per image. The challenge [1] asks for one mask per filament and scores them with PQ [3].

The data has a property that shapes every decision: annotators disagree strongly, and each annotator is scored separately against the same prediction. A prediction can therefore never match all annotators, and the best achievable PQ is far below 1. Our contributions are (i) a simple and reproducible pipeline; (ii) a component scorer that selects instances by their expected PQ contribution, which beats the hand rules we tuned; (iii) a measurement of the annotator ceiling and an error analysis that explains why the common improvements fail; and (iv) the negative results, reported with numbers.

II. Data and metric

The training set has 707 images (2048×2048 px) with 8,199 annotated filaments; 411 images have one annotator, 145 two and 151 three, giving 1,154 annotator-image entries. The test set has 180 images. For each entry, a ground-truth (GT) and a predicted mask form a true positive (TP) when their IoU exceeds 0.5; PQ is $\sum_{TP} \mathrm{IoU}/(|TP| + 0.5|FP| + 0.5|FN|)$, pooled over all entries. It factors into segmentation quality SQ (mean IoU of the TPs) and recognition quality RQ.

Our implementation of the metric reproduces the organisers' self-evaluation notebook exactly. Reading that code showed that it does not enforce one-to-one matching, so two overlapping predictions of one filament would both count as TPs. Duplicating our own confident out-of-fold masks would raise PQ from 0.409 to 0.61 without detecting anything new, since PQ then tends to the SQ of the duplicated masks. We did not submit such files: all our submissions are disjoint by construction.

III. Method

A. Input and target

Each image is area-downsampled to 1536 px and normalised with fixed constants. The training target is the consensus: the fraction of annotators marking each pixel, computed at 2048 px and then downsampled. A network trained with binary cross-entropy (BCE) on this soft target learns the probability that an annotator marks a pixel, which is the quantity the multi-annotator metric rewards.

B. Network and training

A U-Net [4] from segmentation_models_pytorch [6] with an ImageNet ResNet-34 encoder [5], adapted to one input channel (24.4 M parameters). Loss: BCE + soft Dice. AdamW, learning rate $5 \cdot 10^{-4}$ with one epoch of warm-up and cosine decay, weight decay 10^{-4}, batch 4, 40 epochs. Augmentation: flips, 90-degree rotations, affine (±30°, scale 0.9–1.1), brightness, contrast and gamma. The final model is trained on all 707 images (36 min on an NVIDIA A100 via Hugging Face Jobs; about 3 h on an Apple M3 Max).

C. From probabilities to instances

At inference the probability is averaged over four flipped views (identity, horizontal, vertical, both) and upsampled bilinearly to 2048 px. The map is thresholded at 0.45. A morphological closing with a 31 px disk is used only to decide which fragments belong to one filament: the scored masks keep the thresholded pixels. The 8-connected components of at least 100 px are the candidates; they are disjoint by construction.

D. Component scorer

Every false positive costs half a point in the PQ denominator, so the choice of which candidates to submit matters as much as their shape. A hand rule (area ≥ 300 px and mean probability ≥ 0.8) treats all candidates alike. Instead, for each OOF candidate we compute the target $y = \frac{1}{n}\sum_e \mathrm{IoU}_e\,[\mathrm{IoU}_e > 0.5]$ over the n annotator entries e of its image, i.e. its expected matched IoU per entry, and regress it with a gradient-boosted tree model (scikit-learn HistGradientBoostingRegressor) on 20 features: area, the mean, maximum, spread and quantiles of the probability inside, elongation, length, width and compactness, the brightness contrast with a 15 px ring around the component, distance from disk centre, latitude and longitude on a disk fitted with a Hough circle, and two image-level features (number of candidates, mean probability). A candidate pays when y exceeds about PQ/2 (Section VI); we keep candidates whose prediction exceeds $\tau = 0.20$ (PQ varies by less than 0.001 over $\tau = 0.15$–0.25). Inference, including features, takes 4.5 s per image.

E. Validation and tuning

Filaments persist for days, so images from the same week are near-duplicates. We use a 5-fold GroupKFold grouped by calendar month. The threshold and closing are grid-searched with the exact metric on the pooled out-of-fold (OOF) probabilities of the five fold models (707 images). The scorer is cross-validated on the same folds: the model applied to a fold's candidates is trained on the other four. Local OOF PQ ran 0.06–0.07 above the public leaderboard across submissions; every choice was made on OOF, never on the leaderboard.

IV. Results

Table I gives the main numbers. The OOF figures use the submitted pipeline on predictions of models and scorers that never saw the image. To put them in scale, we score each annotator's masks as predictions against every other annotator of the same image (296 images with two or three annotators).

The model exceeds a single annotator on both boundary quality (SQ) and detection (RQ): trained on the consensus, it predicts the filaments most annotators agree on. Fig. 1 shows the three evaluation plots requested by the organisers on the OOF predictions. Over GT/prediction pairs with IoU > 0, the mean IoU is 0.573 and the mean Dice 0.707 (0.553 and 0.688 for the 1024 px rule pipeline). 3.4% of predicted filaments overlap more than one annotation (merges) and 3.7% of annotations overlap more than one prediction (splits); 11.7% of predictions overlap no annotation of their entry and 23.3% of annotations overlap no prediction. Compared with the hand rule, the scorer removes a quarter of the false positives at the cost of more missed annotations, mostly filaments few annotators agree on. The resolution gain is consistent but small: tuned per fold, 1536 px beats 1024 px on four of five folds (+0.009, −0.002, +0.009, +0.005, +0.007), and 2048 px matches 1536 px on the two folds tried. The scorer beats the submitted hand rule (confidence ≥ 0.8) on all five folds, by 0.002–0.009, and the best pooled hand rule (confidence ≥ 0.85, PQ 0.420) by 0.002. Fig. 2 shows typical images.

TABLE I
Main results. OOF: out of fold, 707 images, 1,154 entries.

	PQ	SQ	RQ
Final pipeline, OOF (all 707 images)	0.422	0.674	0.626
Final, OOF (296 multi-annotator images)	0.423	0.671	0.630
Annotator vs. annotator (same images)	0.340	0.634	0.536
1024 px, 2 views, hand rule	0.409	0.666	0.614
+ 4 views	0.412	0.667	0.617
+ component scorer	0.417	0.668	0.624
1536 px, 4 views, hand rule	0.417	0.671	0.621
Public leaderboard, final pipeline	0.35	–	–
Public leaderboard, 1024 px rule	0.35	–	–

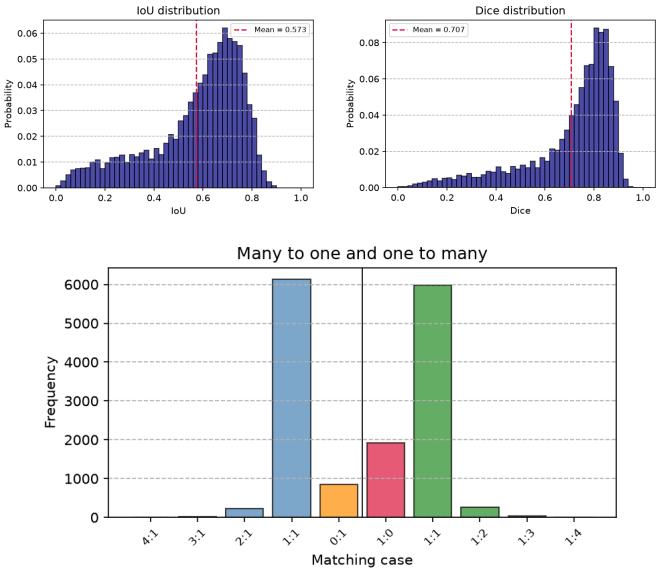


Fig. 1. Out-of-fold IoU and Dice distributions over GT/prediction pairs with IoU > 0, and the one-to-many / many-to-one counts (left: number of GT linked to each prediction; right: number of predictions linked to each GT).

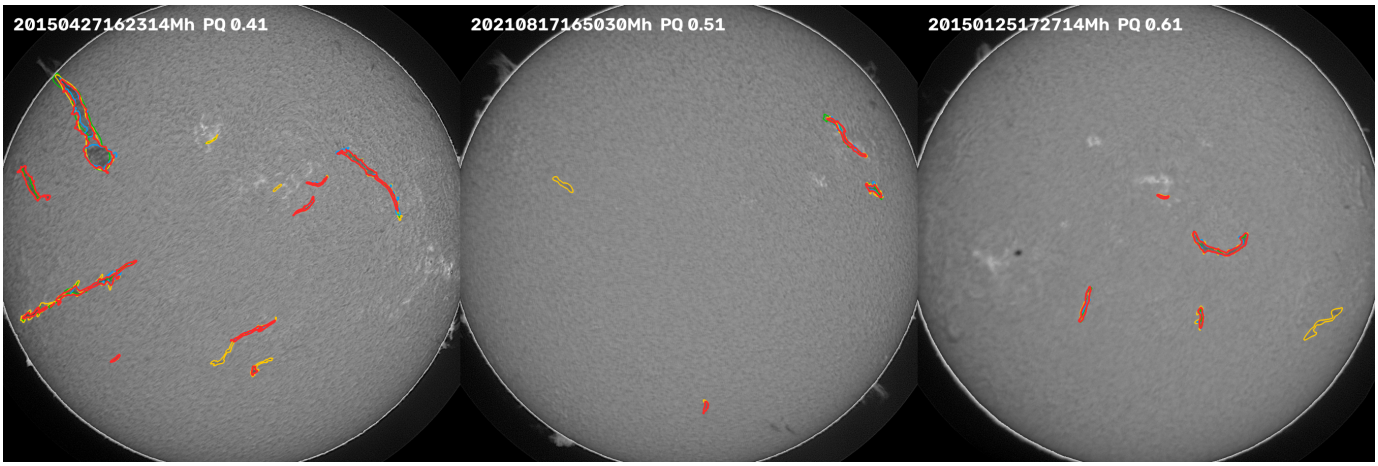


Fig. 2. Out-of-fold predictions (red) on three validation images with three annotators (green, yellow, blue), at the median, upper-quartile and near-best image PQ. Many misses are filaments marked by one annotator only.

TABLE II
Variants (fold 0, 142 images, tuned PQ) and post-processing rules (5-fold OOF, 707 images).

Variant (fold 0)	PQ
U-Net ResNet-34, 40 epochs (baseline)	0.416
10 / 20 epochs	0.389 / 0.396
+ crop refiner at native resolution	0.413
+ Mask R-CNN instances not covered by the U-Net	0.417
Mask R-CNN [7] alone (disjoint masks)	0.377
ResNet-50 encoder	0.414
Limb-darkening-corrected input	0.411
BCE + Lovász hinge [8]	0.390
Average of ResNet-34 and ResNet-50	0.416
Post-processing rule (OOF)	**PQ**
Submission setting (confidence ≥ 0.8)	0.409
Confidence ≥ 0.85 (reference for the rows below)	0.411
No closing, no confidence or limb filter	0.398
Size-dependent confidence threshold	0.411
Hysteresis (seed and grow thresholds)	0.409
8 dihedral views (fold 0; 4 flips: 0.419)	0.419
2048 px, folds 0 / 3 (1536 px: 0.403 / 0.387)	0.404 / 0.388
Masks dilated / eroded by 1 px	0.381 / 0.399
70-degree limb cut-off	no change

V. What did not help

Table II lists the variants we tried, each evaluated with its own tuned post-processing.

Instance models. A Mask R-CNN detector learns grouping explicitly, but its masks are coarser (SQ 0.642 vs. 0.657 for a U-Net trained for a similar number of epochs) and it scores below the U-Net; adding its extra instances or a per-filament crop refiner (following Reche & Cid [9]) changes PQ by less than 0.002 on the fully trained U-Net.

Larger encoder, normalised input, IoU loss. A ResNet-50 encoder, an input divided by its own radial brightness profile (which also removes the scattered-light halo), and a Lovász hinge loss all match or underperform the baseline. With the corrected input the best threshold moves from 0.45 to 0.65: the network becomes more confident, not more accurate.

Ensembles. Averaging two equal-strength models with different encoders gives the same PQ as the better one, and the 5-fold average gives the same leaderboard score as one model. The models share their errors because the errors come from the annotators, so the final submission is one network.

VI. Where the remaining PQ is lost

Near misses. Of the 3,174 OOF false negatives, 45% overlap no prediction, 18% overlap one at IoU < 0.3 and 37% at IoU 0.3–0.5. Among the near misses the prediction is smaller than the GT in 46% of cases and larger in 34%; only 17% are merges with another filament and 16% are splits. Dilating or eroding every mask by a single pixel costs 0.01–0.03 PQ, because filaments are only a few pixels wide: the extent errors are per instance, not a bias a rule could correct.

Small filaments. Recall falls to 0.36 below 500 px. Annotators rarely agree on them: the share of other annotators marking the same filament (IoU > 0.5) is 0.28 below 300 px and about 0.60 above 1,000 px.

When a doubtful prediction pays. A prediction matching k of the n annotators of an image adds about $k\cdot$IoU to the numerator and $0.5n$ to the denominator, so it raises PQ only if $k/n > \mathrm{PQ}/(2\,\mathrm{SQ}) \approx 0.31$. A filament marked by one annotator of three is break-even, which is why lowering the area or confidence thresholds never helped; 28% of our false positives match another annotator's filament on the same image.

VII. Conclusion

A consensus-trained U-Net at 1536 px with a learned component scorer gives OOF PQ 0.422 on MAGFiLO,

above the 0.340 that one annotator scores against another. Choosing instances by their expected contribution to the metric helped more than any change to the segmenter; larger models, instance heads, input normalisation, IoU losses, ensembling and hand post-processing rules did not, and the error analysis attributes most of the remaining loss to disagreement between annotators. Further gains should come from modelling the annotators (e.g. predicting each annotator's style) rather than from a stronger segmenter. Code, weights and a notebook reproducing the submission are in the repository; every experiment is logged in docs/DESIGN.md.

Acknowledgment

This work utilizes GONG data obtained by the NSO Integrated Synoptic Program, managed by the National Solar Observatory, which is operated by the Association of Universities for Research in Astronomy (AURA), Inc. under a cooperative agreement with the National Science Foundation and with contribution from the National Oceanic and Atmospheric Administration. The GONG network of instruments is hosted by the Big Bear Solar Observatory, High Altitude Observatory, Learmonth Solar Observatory, Udaipur Solar Observatory, Instituto de Astrofísica de Canarias, and Cerro Tololo Interamerican Observatory.

References

[1] A. Ahmadzadeh, D. J. Kempton, Q. Li, A. A. Pevtsov, "Solar Filament Segmentation Challenge 2026," Kaggle, 2026, https://www.kaggle.com/competitions/filament-segmentation-2026.

[2] A. Ahmadzadeh et al., "A dataset of manually annotated filaments from H-alpha observations," Scientific Data, 2024, doi:10.1038/s41597-024-03876-y.

[3] A. Kirillov, K. He, R. Girshick, C. Rother, P. Dollár, "Panoptic segmentation," in CVPR, 2019.

[4] O. Ronneberger, P. Fischer, T. Brox, "U-Net: convolutional networks for biomedical image segmentation," in MICCAI, 2015.

[5] K. He, X. Zhang, S. Ren, J. Sun, "Deep residual learning for image recognition," in CVPR, 2016.

[6] P. Iakubovskii, "Segmentation Models Pytorch," GitHub repository, 2019.

[7] K. He, G. Gkioxari, P. Dollár, R. Girshick, "Mask R-CNN," in ICCV, 2017.

[8] M. Berman, A. Rannen Triki, M. B. Blaschko, "The Lovász-softmax loss," in CVPR, 2018.

[9] A. Reche, C. Cid, "New dataset and framework for filament detection, classification, and segmentation using deep learning," Acta Astronautica, vol. 249, pp. 381–398, 2026, doi:10.1016/j.actaastro.2026.07.010.